Opinion Network Formation

Assignment 1: LinkedOut

September 20, 2026

1 Team Name

LinkedOut

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2 GitHub Link to Code

Repository: https://github.com/dawnbliss-coder/opinion-network-formation

3 Dataset Documentation

3.1 Data Description

The dataset comprises survey responses from a class cohort regarding 60 Likert-scale statements spanning four key topics: Technology (T), Education (E), Society/Ethics (S), and Environment (V).

3.2 Encoding and Preprocessing

Standard deviation rescaling (without mean shift) was applied to the raw response matrix. This corrects for the difference between "extreme" and "mild" responders without destroying the genuine uniform agreement signal present across the class. Cosine similarity was used as the primary metric on the rescaled matrix as, unlike full z-scoring, which collapses neutral opinions, it preserves directional agreement effectively.

3.3 Handling Missing Data and Data Quality

The initial dataset contained 96 respondents. Nine respondents with more than 50% missing data (nodes 30, 39, 44, 60, 68, 73, 77, 78, 87) were dropped to ensure data quality, yielding a robust final cohort of **87 respondents**. To prevent highly skewed similarities based on only a handful of shared answers, a strict floor (`MIN_OVERLAP = 5`) was applied to pairwise comparisons: any pairwise similarity computed from fewer than 5 jointly answered statements was discarded (`np.nan`).

4 Pipeline Followed

4.1 User–User Network

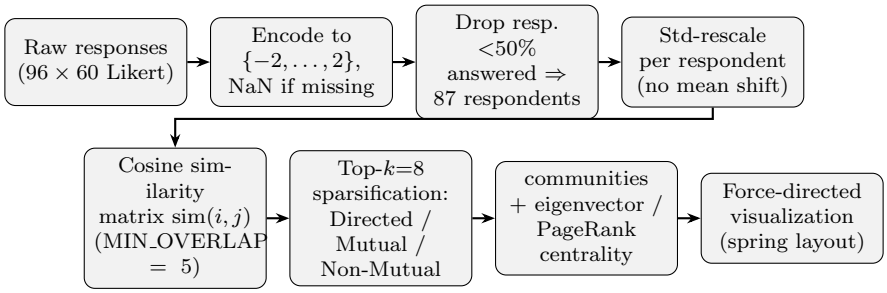


Figure 1: User–User network construction pipeline.

Because survey similarities naturally vary in strength and reciprocity, the User–User network was modeled using a deliberately designed **three-network framework**. Instead of treating one graph as universally "correct," three purpose-built views of the same underlying similarity matrix were generated, each using **top-k sparsification (k = 8)**, chosen because it produces a stable, fully connected base structure in the directed case without artificially isolating nodes.

Directed Top-8 Network (Primary Network). Edges are directed from user A to user B if B is in A's top 8 most similar peers. This network acts as the base representation: cosine similarity's asymmetric top-k membership is a genuine, informative feature of the data, capturing "who looks up to whom" in opinion space. It preserves the full 694 edges and is used for PageRank, in/out-strength, and connected-component metrics.

Mutual Top-8 Network (High-Confidence Projection). Edges exist only if A and B mutually rank each other in their top 8. This projection isolates reciprocal agreement, purposely shattering the network to reveal the deeply connected "consensus core" (the largest component) versus the fragmented periphery.

Non-Mutual Undirected Top-8 Network (Comparison Baseline). An edge exists if *either* A or B is in the other's top 8. This network is used specifically for cross-category and engagement-weighted comparisons. Enforcing mutuality on thinner data subsets (like specific topics) fragments the graph unevenly, so the non-mutual approach guarantees a consistently connected structure necessary for fair modularity and community comparisons.

Methodological implementations: Louvain community detection was used; for the primary directed graph, edge weights for reciprocal pairs were summed (weight-summed symmetrization) rather than dropped, preserving the ∼78% of network structure that is inherently one-directional. NetworkX's Eigenvector Centrality `max_iter` was raised to 1000 to ensure convergence on sparse structures. For the signed network, raw similarity is heavily right-skewed, so thresholds were dynamically calibrated to the extreme tails of the distribution: positive threshold = 0.9 (exceptional similarity) and negative threshold = −0.1 (exceptional dissimilarity).

4.2 Statement–Statement Network

This network flips the viewpoint of the User–User graph: instead of asking which *respondents* answered alike, it asks which *statements* are answered alike by the class. It starts from the same

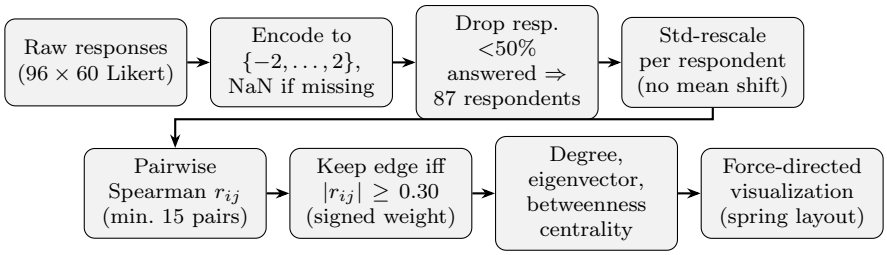


Figure 2: Statement–Statement network construction pipeline.

encoded, engagement-filtered, per-respondent std-rescaled matrix X of size 87×60 (Section 3), with missing entries kept as NaN (`src/statement_network.py`).

1. **Nodes.** One node per statement (60 in total), annotated with its T/E/S/V category and full text.
2. **Similarity.** For every pair of statements (i, j) we compute the Spearman rank correlation r_{ij} between their two columns of X, using only the respondents who answered both (pairwise-complete). Pairs with fewer than 15 overlapping respondents (`MIN_PERIODS = 15`) get no correlation. Spearman is used, rather than the cosine similarity of the User–User network, because the responses are ordinal and here we measure co-variation across respondents. It is computed on the *rescaled* matrix because otherwise "extreme" responders (who mark almost everything Strongly Agree/Disagree) push every pair of columns towards the same corners and inflate the correlation between otherwise unrelated statements.
3. **Edges.** An undirected edge $\{i, j\}$ is kept iff $|r_{ij}| \geq \tau = 0.30$ (`EDGE_THRESHOLD`), with weight $w_{ij} = r_{ij}$ so that the sign is preserved (positive = statements endorsed together, negative = opposing opinions).
4. **Metrics.** *Degree* counts neighbours (unweighted). *Eigenvector centrality* is computed on $|w_{ij}|$, since it requires non-negative weights (`max_iter` = 1000). *Betweenness* is computed on the distance $d_{ij} = 1 - |r_{ij}|$, because NetworkX reads edge weights as traversal costs, and using r directly would make a strong tie look "farther" than a weak one. Connectivity diagnostics (components, largest component, isolated statements) are also recorded, since at higher thresholds statements can silently lose all their edges.

Threshold choice. The cutoff τ controls how dense the network is:

τ	Edges	Density	Negative edges
0.10	1,546	0.8734	31
0.20	1,211	0.6842	5
0.25	1,021	0.5768	3
0.30	**810**	**0.4576**	**0**

At $\tau = 0.10$ the graph is almost complete (87.3% of all pairs), so it barely differentiates between statements. $\tau = 0.30$ gives a more selective network while still retaining one large connected structure, and was used as the final network. Note that this choice also removes every negative edge.

4.3 User–Statement Network

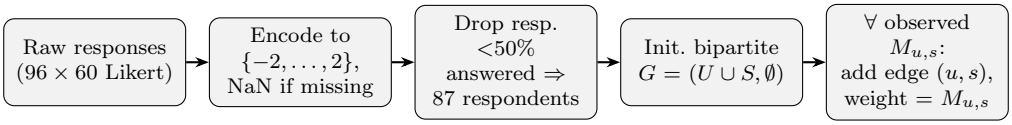


Figure 3: User–Statement network construction pipeline.

Formally, let $M \in \{-2, -1, 0, 1, 2, \text{NaN}\}^{U \times S}$ be the encoded, engagement-filtered response matrix ($|U|$=87 respondents, $|S|$=60 statements). The network is the bipartite graph $G = (U \cup S, E)$ with $(u, s) \in E \iff M_{u,s}$ is observed (not NaN), and edge weight $w_{u,s} = M_{u,s} \in \{-2, -1, 0, 1, 2\}$. No edge is created for a skipped/"No Comments" entry, since skipping $\neq$ a Neutral (0) response.

A reference check (`edge_threshold_table()` in `src/user_statement_network.py`) confirms that sparsifying by response magnitude does not produce a meaningfully sparser graph:

Cutoff	Edges	Density
All answered (incl. Neutral)	5,158	98.8%
Drop Neutral only	4,491	86.0%
Strongly Agree/Disagree only	2,349	45.0%

5 Analysis and Visualizations

5.1 User–User Network

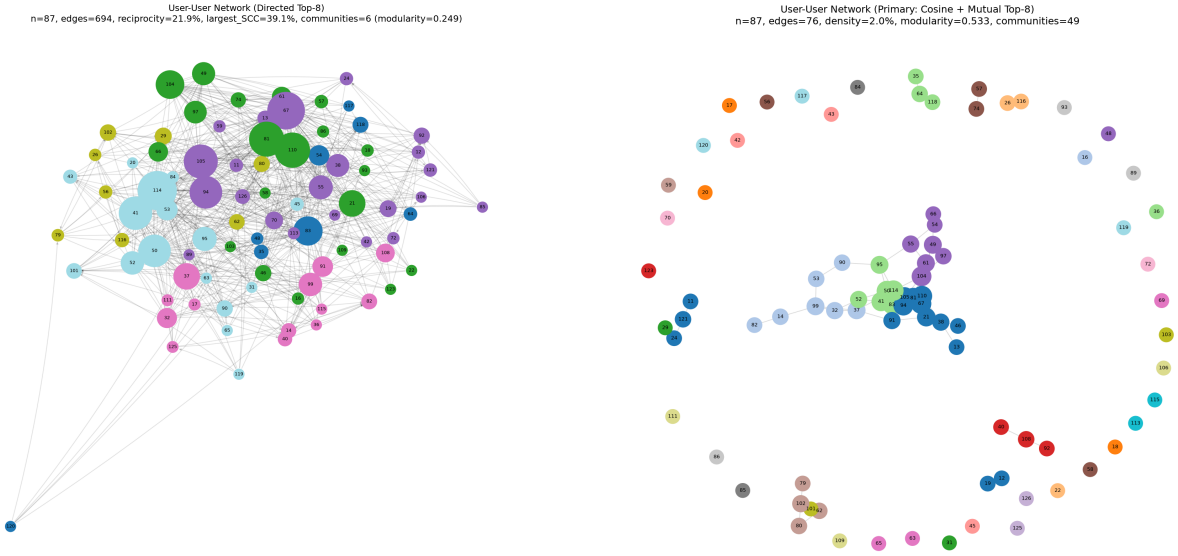


Figure 4: *Left:* Full Directed graph of top 8 similarity scores per user based on cosine similarity. *Right:* Full graph of mutual top 8 similarity scores per user based on cosine similarity.

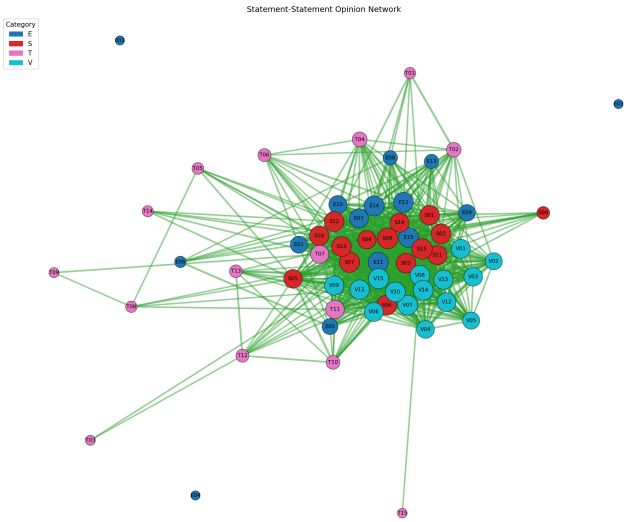
Topic-Specific Multiplex Layers. To analyze how opinions cluster by topic without artificial fragmentation, the non-mutual sparsification was applied to the four distinct statement categories.

Category	Edges	Modularity	Communities	Largest Component
E (Education)	519	0.370	7	86 / 87
S (Society/Ethics)	621	0.313	4	87 / 87
T (Technology)	548	0.377	5	87 / 87
V (Environment)	609	0.299	7	86 / 87

User-User Response Similarity Networks (Non-Mutual). To ensure findings were robust regardless of the specific similarity metric, the baseline cosine similarity graph was compared against alternative network calculations.

Metric	Edges	Modularity	Avg. Clustering	Largest CC
Cosine (Weighted)	625	0.242	0.111	87 / 87
Pearson (Unweighted)	559	0.291	0.196	87 / 87
Spearman (Unweighted)	517	0.302	0.148	87 / 87
Agreement Ratio (Raw Data)	542	0.396	0.243	87 / 87

5.2 Statement–Statement Network



Metric	Value
Nodes / edges	60 / 810
Density	0.4576
Average degree	27.00
Pos. / neg. edges	810 / 0
Connected components	4
Largest component	57 / 60
Isolated nodes	E02, E03, E04

Degree	Eigenvector	Betweenness
S09 (45)	S09 (0.1993)	S15 (0.0479)
S15 (44)	V07 (0.1907)	V09 (0.0421)
S07 (43)	V14 (0.1845)	T07 (0.0374)
E11 (42)	S15 (0.1842)	S07 (0.0368)
E14 (42)	V15 (0.1814)	S13 (0.0362)

Figure 5: Statement–Statement opinion network ($|r| \geq 0.30$). *Left:* node colour = T/E/S/V category, node size $\propto$ degree, edge width $\propto |r|$; all edges are positive (green) since no negative correlation reaches the threshold. *Right:* network summary and the top five statements under each centrality measure, ranked (value in brackets; the first letter of each ID is its category).

Figure 5 shows that the network is dense (810 edges among 60 statements) and dominated by one connected component, with E02, E03 and E04 left without edges. The three centrality measures single out different statements (S09, S15, V07 and T07 recur), which Section 6.2 discusses.

5.3 User–Statement Network

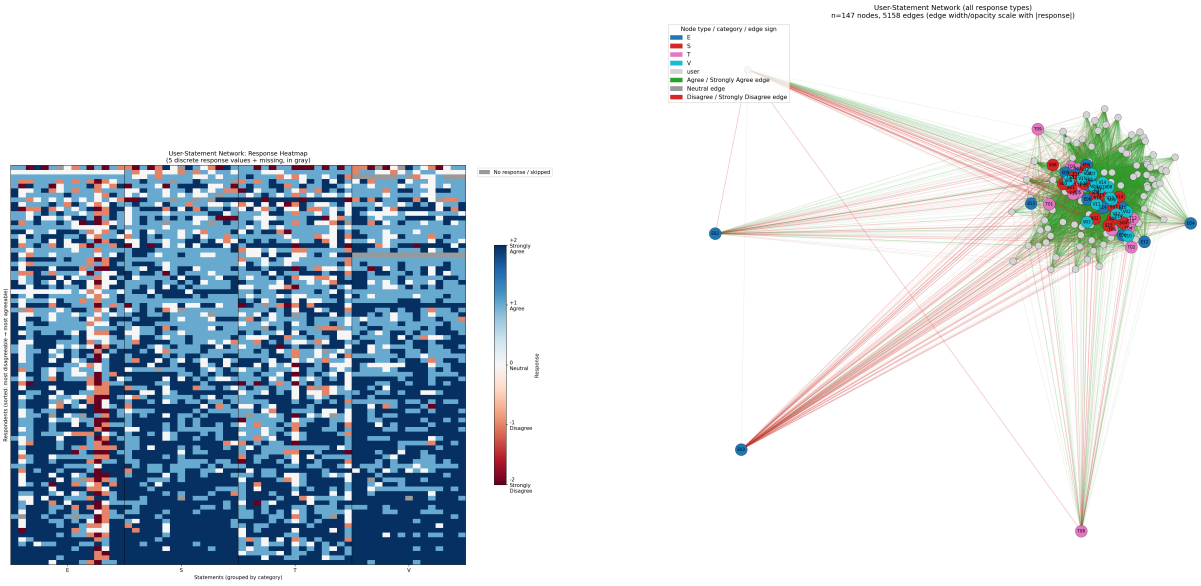


Figure 6: *Left:* response heatmap (respondents sorted most disagreeable → most agreeable; statements grouped by category; 5 response colors + gray for missing). *Right:* full bipartite network, all 5,158 edges and all five response types drawn (green = agree, red = disagree, gray = neutral; width/opacity $\propto$ |weight|).

Metric	**Value**
Users / Statements	87 / 60
Edges (answered)	5,158 / 5,220 possible
Density $= \lvert E\rvert/(\lvert U\rvert\lvert S\rvert)$	98.8%
Mean edge weight $= \frac{1}{\lvert E\rvert}\sum_{(u,s)\in E} w_{u,s}$	+1.14
Response split	80.0% positive, 12.9% neutral, 7.1% negative

Density near 100% means missingness does not carry much structure here. Almost everyone answered almost everything, so the real signal lives in edge *weight* rather than edge presence. The positive mean weight shows that the class leans toward agreement across the survey as a whole.

Per-node metrics: let $N(u) = \{s : (u, s) \in E\}$ (statements u answered) and $N(s) = \{u : (u, s) \in E\}$ (respondents who answered s), so $\deg(u) = |N(u)|$, $\deg(s) = |N(s)|$.

Metric	Formula	What it tells us
mean_response(u)	$\frac{1}{\lvert N(u)\rvert}\sum_{s\in N(u)} w_{u,s}$	Respondent u's average stance. Used to rank who is most agreeable and who is most disagreeable overall.
mean_abs_response(u)	$\frac{1}{\lvert N(u)\rvert}\sum_{s\in N(u)} \lvert w_{u,s}\rvert$	Respondent u's opinion *intensity*, independent of direction. This separates decisive respondents from wishy-washy ones (someone alternating between Strongly Agree and Strongly Disagree scores high here even if their mean_response is close to 0).
mean_response(s)	$\frac{1}{\lvert N(s)\rvert}\sum_{u\in N(s)} w_{u,s}$	Statement s's consensus level and direction (near ± 2 means broad agreement or disagreement, near 0 means the respondents are split).
std_response(s)	$\sqrt{\frac{1}{\lvert N(s)\rvert - 1}\sum_{u\in N(s)} (w_{u,s} - \bar{w}_s)^2}$	Spread of opinion on s. A low std together with a high |mean| points to a consensus statement, while a high std points to a polarizing or contested statement, regardless of its average.

For respondents, the extremes by these metrics are:

Type	Respondent	mean_response	mean_abs_response
Most agreeable	41	1.77	1.87
Most agreeable	94	1.75	1.82
Most disagreeable	120	−0.21	1.29
Most disagreeable	119	0.27	0.47

Respondent 120 is the only respondent whose average stance leans toward disagree overall. For statements, the strongest consensus and most polarizing cases by these metrics are:

Type	Statement (category)	mean_response	std_response
Consensus	E15 (E)	1.71	0.50
Consensus	V13 (V)	1.67	0.56
Polarizing	E04 (E)	0.44	1.17
Polarizing	E03 (E)	−0.94	1.12

All three most polarizing statements (E02, E03, E04) come from the Education category.

6 Results and Discussion

6.1 User–User Network

A dominant consensus core. A small, highly representative cluster of respondents acts as the ideological center of the class. Respondents 81, 105, 110, 114, and 67 consistently top the charts across both in-strength (most nominated as similar) and PageRank in the Directed network. Nodes 81, 105, 94, 67, and 110 also top eigenvector centrality in the Mutual network. They represent the "class average," acting as gravitational hubs in the opinion space.

Hub-and-spoke agreement (low reciprocity). Similarity in the class is fundamentally directional. Only 21.9% of directed edges are reciprocated. While the entire class is weakly connected (100% WCC), the largest strongly connected component (SCC) is just 39% of nodes (34 nodes). Most students "point to" the consensus core, but the core does not point back. When mutuality is strictly enforced, the network shatters into 49 micro-communities with 57 unconnected nodes, showing that broad opinion camps are held together by mostly one-directional similarity.

Polarization exists only at the extreme margins. Using the strictly calibrated signed network (thresholds 0.9 and -0.1), the extreme tails of agreement and disagreement are relatively balanced but incredibly sparse: 60 positive edges vs. 47 negative edges out of thousands of possible pairs. Deep ideological opposites do exist in the class, but they are rare, disagreement overall is moderate, there are no two big heavily polarized opinion groups.

Opinion outliers. Looking at bottom out-strength scores in the Directed graph identifies individuals whose closest peers still aren't very similar to them. Respondent 120 (out-strength 0.23) is a massive anomaly, scoring an order of magnitude lower than the rest of the cohort (the next lowest is node 119 at 3.00). This respondent's response pattern is highly idiosyncratic compared to the broader class consensus.

Topic-driven factionalism. While the class shows general consensus, factionalism spikes depending on the subject matter. Topics related to Technology (modularity 0.377, 5 communities) and Education (modularity 0.370, 7 communities) display significantly higher modularity, meaning tighter, more distinct echo chambers, than Environment (0.299) or Society (0.313). This suggests that Tech and Education statements are where the most distinct dividing lines in class opinion are drawn.

6.2 Statement–Statement Network

A dense, uniformly positive structure. At $|r| \geq 0.30$, 810 of the 1,770 possible statement pairs (45.8%) are linked (average degree 27 out of 59), and *every* retained edge is positive. Answers to many different statements therefore move together across respondents, which points to substantial shared structure in the class's opinions rather than 60 independent views. This does not mean the class never disagrees: weaker negative correlations exist (31 negative edges at $|r| \geq 0.10$, 5 at 0.20, 3 at 0.25), but none reaches the 0.30 threshold.

One broad component and three detached statements. 57 of the 60 statements (95%) lie in a single connected component, so the network is not split into separate, equally sized groups of statements. The remaining three are isolated, i.e. none correlates with any other statement at $|r| \geq 0.30$: E02 (written exams measure knowledge accurately), E03 (compulsory attendance) and E04 (online learning complementing classroom teaching). These are exactly the three statements found to be the most contested in the User–Statement analysis (highest std, Section 5.3). This suggests that where the class divides on these Education issues is largely unrelated to how respondents answer the rest of the survey.

Hubs and bridges. S09 (universities should promote an inclusive, respectful environment for differing viewpoints) has the highest degree (45 of a possible 59) and the highest eigenvector centrality (0.1993): it co-varies with many statements, and those are themselves well connected. S15 (public trust is essential for responsible adoption of emerging technologies) is the only statement in the top five on all three measures: second in degree (44), fourth in eigenvector centrality (0.1842) and first in betweenness (0.0479), making it both a hub and the strongest bridge between regions of the

network. Degree and betweenness are not interchangeable: T07 (robots replacing humans in hazardous jobs) is outside the degree top five yet third in betweenness, so its importance comes from where it sits rather than from its number of links. That S09 is also one of the strongest consensus statements (Section 6.3) shows that centrality here reflects co-variation with other statements, not high average agreement.

Structure cuts across the survey's categories. The top-ranked statements come from all four categories (S, E, V, T) rather than from one: for example, the Environment statements V07, V14 and V15 sit alongside the Society statements S09 and S15 among the most central nodes. Response patterns therefore link statements across the survey's own T/E/S/V boundaries. This is evidence of cross-category association in how respondents answered, not of the original categories being wrong.

Robustness and caveats. S09, S15, V07, V14 and V15 repeatedly appear among the most central statements across the thresholds tried, so their prominence does not depend entirely on the single choice $\tau = 0.30$, although all findings remain conditional on it. The network describes associations between response patterns, not causal influence: a highly connected or high-betweenness statement is not necessarily one the class strongly agrees with, nor one that drives opinions on other issues.

6.3 User–Statement Network

The near-total density (98.8%) combined with a positive mean edge weight (+1.14) shows the class leans broadly toward agreement across the survey, consistent with the 80%/13%/7% positive/neutral/negative split (Section 5.3): most respondents answered most statements, and most answers were Agree or Strongly Agree.

At the respondent level, mean_response isolates a small set of genuine outliers from the agreeable majority. Respondent 120 has the lowest mean_response (-0.21), the only respondent whose average stance falls on the disagreement side of neutral despite answering 56 of 60 statements. That looks like a real dissenting voice rather than a low-engagement artifact. Notably, this is the same respondent (120) flagged as the primary out-strength outlier in the User–User network (Section 6.1), corroborating across networks that respondent 120 is a genuine opinion outlier rather than an artifact of either method.

At the statement level, std_response identifies the Education-category statements E02, E03, and E04 as the most contested in the entire survey, while E15 and V13 are the strongest consensus statements. It's worth pointing out that **S09**, the most structurally central statement in the Statement–Statement network (highest degree and eigenvector centrality), also shows up among the most consensus statements here (mean 1.47, std 0.59). So S09's centrality seems to come from *broad shared agreement* with many other statements rather than from controversy: centrality and polarization are measuring two different things.

7 Individual Contribution

Member	Contribution
Shreyas Agarwal	User–User network: three-network framework (Directed/Mutual/Non-Mutual Top-8 sparsification), community detection, eigenvector and PageRank centrality, signed-network threshold calibration, topic-layer and alternative-metric comparisons, corresponding report sections.
Sarah Roomi	Statement–Statement network: pairwise Spearman correlation matrix on the rescaled responses (`statement_network.py`), thresholded signed/weighted edge construction ($\|r\| \geq 0.30$) with a threshold sweep (0.10–0.30), degree / eigenvector / betweenness centrality and connectivity diagnostics, force-directed visualization, corresponding report sections.
Priyanka Agarwal	User–Statement network: bipartite graph construction (`user_statement_network.py`), response-weighted edge scheme, per-user/per-statement metrics, heatmap and full-network visualizations, corresponding report sections.